

Python Practice Questions Compilation

This document contains topic-wise Python practice questions for Python Basics.

Operators

1. Write a program to take two numbers from the user and perform all arithmetic operations.
2. Create a calculator using +, -, *, /, %, //, and ** operators.
3. Compare two numbers using comparison operators.
4. Check whether a number is between 50 and 100 using logical operators.
5. Demonstrate identity and membership operators using lists.

If Else

1. Check whether a number is even or odd.
2. Take marks and print grade according to marks.
3. Check whether a year is leap year or not.
4. Create a login system using if else.
5. Find the largest among three numbers.

Loops

1. Use nested loops to print a rectangle star pattern.
2. Print multiplication tables using nested loops.
3. Print a triangle number pattern using nested loops.
4. Count vowels in a string using loops.

Lists

1. Create a list of numbers and print only even numbers.
2. Add, remove, and update elements in a list.
3. Store 5 student names in a list.
4. Find maximum and minimum value in a list.
5. Merge two lists.

Tuples

1. Create a tuple of cities and print them using loops.
2. Count occurrences of a value in a tuple.
3. Create a tuple with mixed data types.
4. Convert tuple into list and modify it.
5. Concatenate two tuples.

Sets

1. Find union and intersection of two sets.
2. Remove duplicates from a list using sets.
3. Store names in a set.
4. Check whether a value exists in a set.
5. Print odd values from a set.

Strings and Slicing

1. Print first and last character of a string.
2. Reverse a string using slicing.
3. Count total words in a sentence.
4. Extract middle character of a string using slicing.
5. Check whether a string is palindrome.

String Functions

1. Remove extra spaces using strip().
2. Split a sentence using split().
3. Replace a word using replace().
4. Count occurrences of a word in a string.
5. Convert string into uppercase and lowercase.

Dictionaries

Example

```
student = {  
    "name": "Ali",
```

```
"subjects": ["Math", "Physics", "Computer"]
}
```

```
print(student["subjects"][0])
```

1. Create a student dictionary with name, age, and marks.
2. Update values inside a dictionary.
3. Store a list inside a dictionary key and access it.
4. Loop through keys and values of a dictionary.
5. Create a product-price dictionary and find highest price.

Question Shopping Cart and Login Management System

Create a Python program for a **Shopping Cart and Login System** using all basic Python concepts.

Requirements:

1. Create a login system using:
 - Dictionary for usernames and passwords
 - If-else for validation
 - Loops for multiple login attempts
2. After login:
 - Show a shopping menu
 - Store products and prices in a dictionary
3. Allow users to:
 - Add products to cart
 - Remove products
 - Update quantities
4. Store cart items inside lists and dictionaries.
5. Use loops to calculate:
 - Total bill
 - Quantity of products

- Most expensive product
- 6. Use sets to display unique purchased products.
- 7. Use nested loops to print:
 - Product patterns
 - Bill design using stars
- 8. Use string functions:
 - `split()` for product input
 - `replace()` for invoice formatting
 - `strip()` for cleaning user input
- 9. Display:
 - Final bill
 - Discount using operators
 - Customer status using if-else conditions